

University of Dhaka
Affiliated Engineering Colleges
Department of Computer Science and Engineering
2nd Year 1st Semester B.Sc.(Engg.) Final Examination-2022

EEE 2104 : Electronic Devices and Circuits

Time: 3 Hours

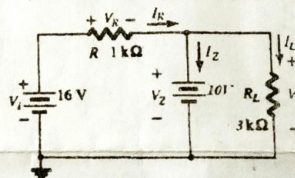
Total Marks: 70

Answer any 5 (Five) set of the following Questions

1. a) What is semiconductor? Why do we use semiconductors? 1+2 *
 b) Differentiate between extrinsic and intrinsic semiconductor. 3
 c) Explain the volt-ampere (V-I) characteristics of PN junction. 5
 d) What is doping? Why it is done? 1+2

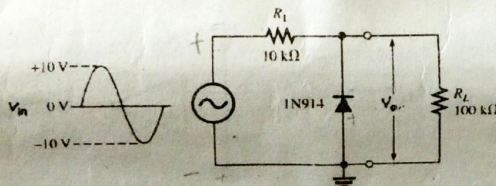
2. a) Explain working principle of a full wave bridge rectifier with a neat sketch. 4
 b) What is LED? Explain the working principle of LED. 1+3
 c) Write down advantage and disadvantage of full wave rectifier. 03
 d) An a.c supply of 230 V is applied to a half wave rectifier circuit through a transformer of turn ratio 10: 1 Find (i) the output d.c voltage and (ii) the peak inverse voltage. Assume the diode to be ideal. 3

3. a) What is Zener diode? Explain how does zener diode regulate voltage? 1+4
 b) For the Zener diode regulator, Determine: (i) V_L (ii) V_R (iii) I_Z (iv) P_Z 5

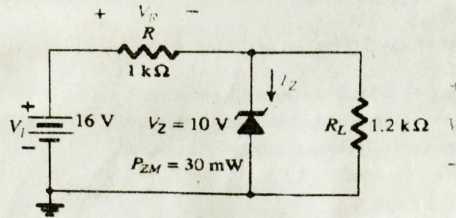


- c) Discuss working principle of JFET with necessary diagram. 4

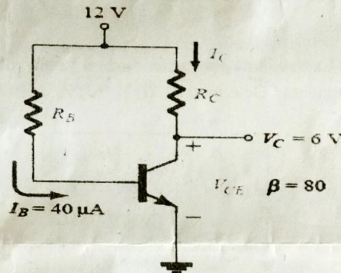
4. a) How does transistor work as a switch? 3
 b) Why does the transistor need biasing? 3
 c) What is stability factor? For a transistor, Prove that $\beta = \frac{\alpha}{1-\alpha}$. 1+3
 d) What is clipper circuit? Determine the output waveform for network of the following figure [Assume the diode to be Silicon]. 1+3



5. a) What is SCR? Explain operating principle of SCR. 05
 b) What is TRIAC? Explain operating principle of TRIAC. 05
 c) For the following Zener diode network, determine V_L , V_R , I_Z , and P_Z . 04



6. a) What is FET? Describe the operating principle of D-MOSFET. 06
 b) Discuss the operation of summing amplifier. 03
 c) Determine R_B , R_C , I_C , V_B and V_{CE} for the following fixed-bias configuration. 05



7. a) What is the importance of power electronics? 2
 b) What is non-inverting amplifier? Derive the output voltage equation of non-inverting amplifier 05
 c) What is an op-amp? What is the need of negative feedback in an op-amp? 1+2
 d) Determine the output voltage for the circuit of Figure with a sinusoidal input of 2.5 mV. Where $R_f = 10\text{ K}\Omega$ and $R_{in} = 470\text{ }\Omega$. 04

